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Chapter 6

PELVIC SHUNTS

Claude Franceschi
Saint Joseph and Pitié-Salpétrière Hospitals, Paris, France.

1. PELVIC ESCAPE POINTS

Pelvic escape points (PEP) represent an important source of varicose veins [161]. They are defined according to their location and the tributaries of the hypogastric veins that feed them. Superior and inferior gluteal veins feed the superior and inferior gluteal points (SGP and IGP) at the buttocks; the obturator vein feeds the obturatory point (OP) at the groin; the round ligament vein feeds the I point (IP) at the superficial inguinal orifice; and the internal pudendal vein feeds the P point (PP) at the perforator of perineal location. Pelvic veins on one side of the body communicate, vertically and horizontally, with the corresponding veins on the opposite side through numerous plexuses. That is why a closure of one or more pelvic veins cannot completely correct the refluxes at PEPs. In the same way, PEPs are connected to any vein at the same side and at the opposite side. Consequently, any varice of the lower limbs can be due to any escape point, at the same or the opposite side.

1.1. I Point (IP) and P Point (PP)

They come out much more specifically in women during and after pregnancy. The reasons are the excesses of venous pressure due to fistulae effect of the placenta, pelvic vein compression by the voluminous uterus, and the hormonal status that modifies venous compliance. Usually, they are associated with intra-pelvic varicose veins, such as varicocele of broad ligaments. This association is not necessarily one of cause and effect. On the one hand, pelvic varicose veins after pregnancy are virtually constant, but varices of the lower limbs are not. On the other hand, retrograde phlebography in pelvic varicese does not show IP or PP constantly, and the embolization of pelvic varices does not stop the escapes through IP and PP [161]. These points are not precisely identified by phlebography but rather by duplex-scan [106].

1.2. I Point (IP)

IP escape in women consists of the re-opening of the embryological duct of Nuck during pregnancy. This duct re-opening links the incompetent vein of the round ligament of the uterus (equivalent to sinus pampiniformis in males) with the tributaries of the saphenous arch and leads to closed shunts 4 and 5. As several anastomoses connect in particular the external pudendal vein not only with ipsilateral tributaries of the great saphenous arch, but also with controlateral tributaries, varices of the right limb can be due to left I point and vice versa. IP can also feed vulvar varices through connections with labial veins, which are tributaries of the perineal vein (Figure 6.2a,b). IP is located about 2cm above the arch of the great saphenous vein, medially to the the epigastric vein (Figure 6.1). The diagnosis is not constantly reliable with phlebography. Color Doppler in the standing position shows a characteristic backflow during VM. In males, reflux through the IP corresponds to varicocele outflow.

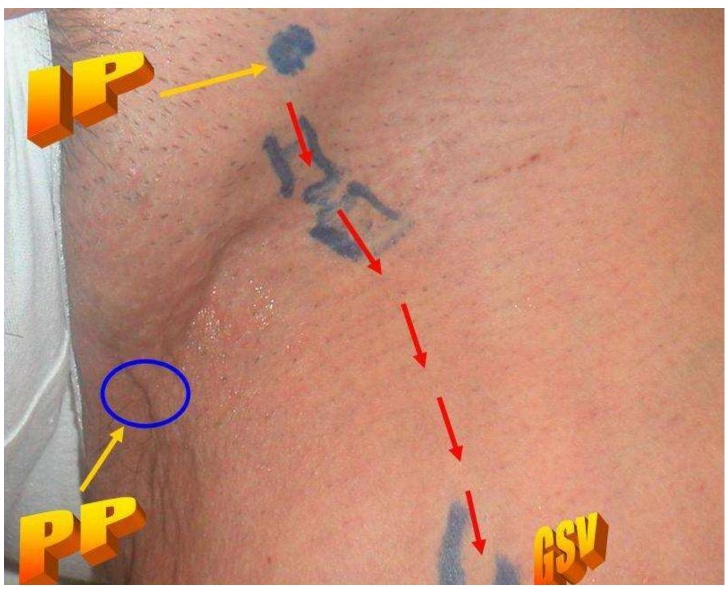

Figure 6.1. IP and PP pre-operative mapping.

1.3. P Point (PP)

PP escape in the perineal vein of women receives labial veins than perforates the superficial perineal fascia at a P point that is located at the posterior edge of the transversal muscle of the perineum. PP skin projection is at the junction of the posterior fourth and the anterior fourth of the external edge of the labia major. PP can feed vulval varices, perineal varices, and closed shunts 4 and 5. Due to transversal anastomosis, it can also feed the same varices on the opposite side (Figure 6.3a,b). As for IP, PP is not always evident in retrograde pelvic phlebography, but can be diagnosed reliably by duplex-scan, which shows reflux at PP during VM [106]. The standing position is not easy for duplex-scan of PP. The best position is gynecologic. The investigation must be superficial and not intra-vaginal. A superficial probe, preferably equipped with a high resolution probe (10 Mhz or more), or with a water interface filing a surgical glove is simply put on the skin of the perineum.

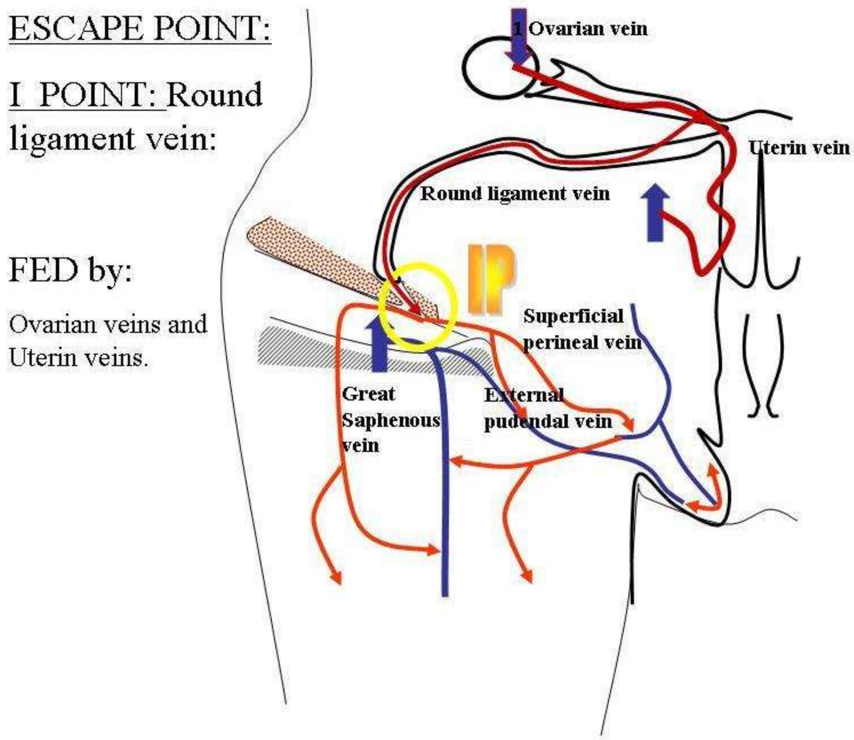

Figure 6.2a. Veins Anastomosis leading to IP.

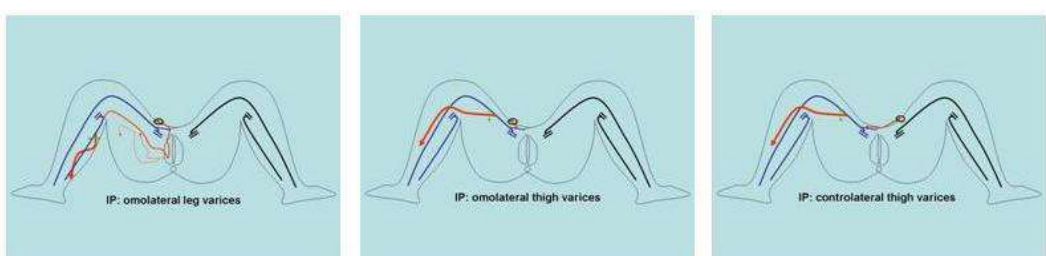

Figure 6.2b. Varices of the limb fed by pelvic shunts linked to IP.

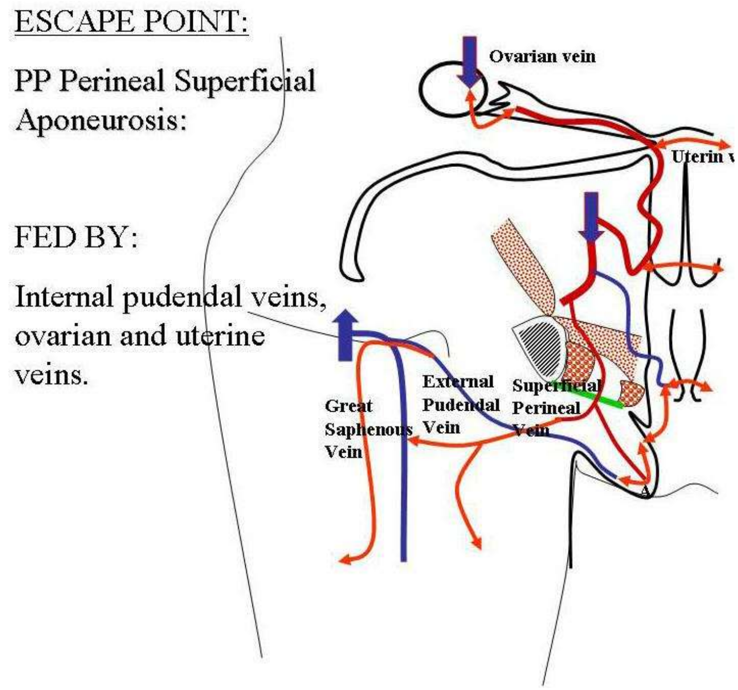

Figure 6.3a. Veins Anastomosis leading to PP.

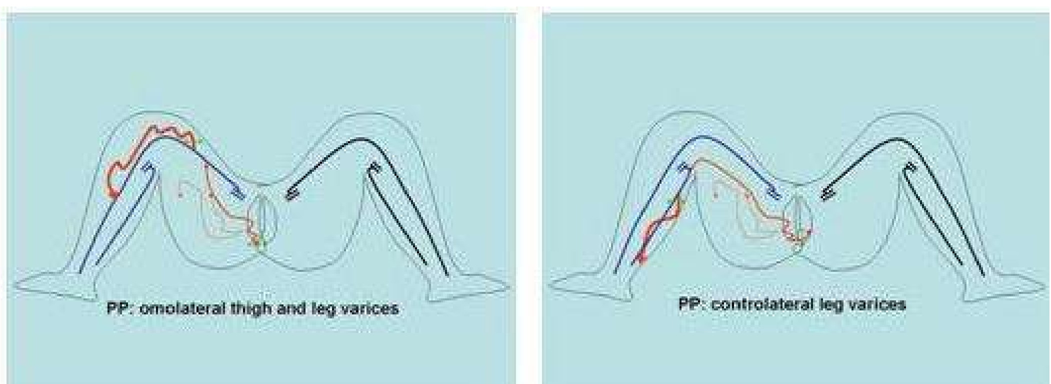

Figure 6.3b. Varices of the limb fed by pelvic shunts linked to PP.

1.4. Obturator Escape Point

The oburator vein (OV) emerges in the thigh through the obturator orifice and usually connects with the common femoral vein. Thus, in the majority of cases, a possible reflux is absorbed by a competent deep network, with neither varices nor clinical symptoms in the limb. Sometimes, however, the OV connects with the arch of the great saphena, in which case reflux in OV can involve closed shunts 4 or 5.

1.5. Superior Gluteal Escape Point (SGEP)

SGEP usually comes out at the middle of the buttock and connects congenital varices of the persistent external vein and/or the arch of the small saphena. It sometimes causes a closed shunt 3 or 5 that involves the small saphena.

1.6. Inferior Gluteal Escape Point (IGEP)

IGEP also occurs, particularly in congenital venous malformations. It usually emerges either at the inferior edge of the buttock or at the popliteal fossa when it feeds varices of the sciatic nerve.

1.7. Venous Shunts in Congenital Venous Malformations (CVM)

Usually, CVM of lower limbs make different presentations under different hemodynamic conditions. Shunts are either VOS, in cases of more or less extended deep venous aplasia, or CS, particularly when embryologic veins persist after birth. In some cases, both VOS and CS can be associated in mixed shunts. These mixed shunts can be overloaded not only by the shunting flow, but also by cutaneous malformations, such as superficial angiomas.

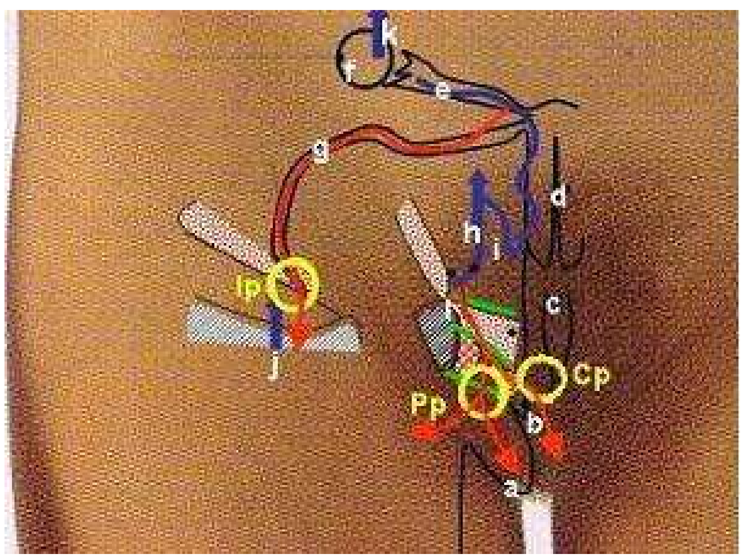

Figure 6.4 A) Visceral pelvic escape points and feeding veins. a: labium major. b:labium minor + clitoris. c: rectum. d: uterus. e: fallopian tube + tubo-ovarian vein. f: ovary. g:round ligament of uterus and its vein. h: hypogastric vein. i: uterine vein. j: great saphena h: hypogastric vein. k: ovarian vein. l: internal pudendal vein. Ip: Inguinal point: reflux location at the superficial inguinal ring is due to round ligament vein (g) incompetence fed by one more out of the visceral tributaries (I ,e) of the hypogastric vein(h) and/or ovarian vein (k). This reflux can feed superficial varices anywhere in the ipsi-lateral limb but in the opposite too either in the labium major (a) through superficial connections with superficial tributaries of the internal pudendal vein (l). Pp: Perineal point: reflux location at the

superficial perineal fascia perforation at the posterior edge transversum muscle, in correspondence with the fourth posterior part of the external edge of the labium major, due to the perineal vein incompetence fed by the internal pudendal vein (I) of the hypogastric vein(h) and/or ovarian vein (k). This reflux can feed superficial varices anywhere in the ipsi-lateral limb but in the opposite too either in the labium major (a) through its labial posterior tributary. Cp: Clitoris point: reflux location at the superficial perineal fascia perforatio next to the proximal clitoris, due to the incompetence of the superficial vein of the clitoris fed by the internal pudendal vein (I) of the hypogastric vein(h) through the incompetent bulbo-clitoridis vein. This reflux can feed superficial varices anywhere in the ipsi-lateral limb but in the opposite too either in the labium major (a) through its labial anterior tributary.

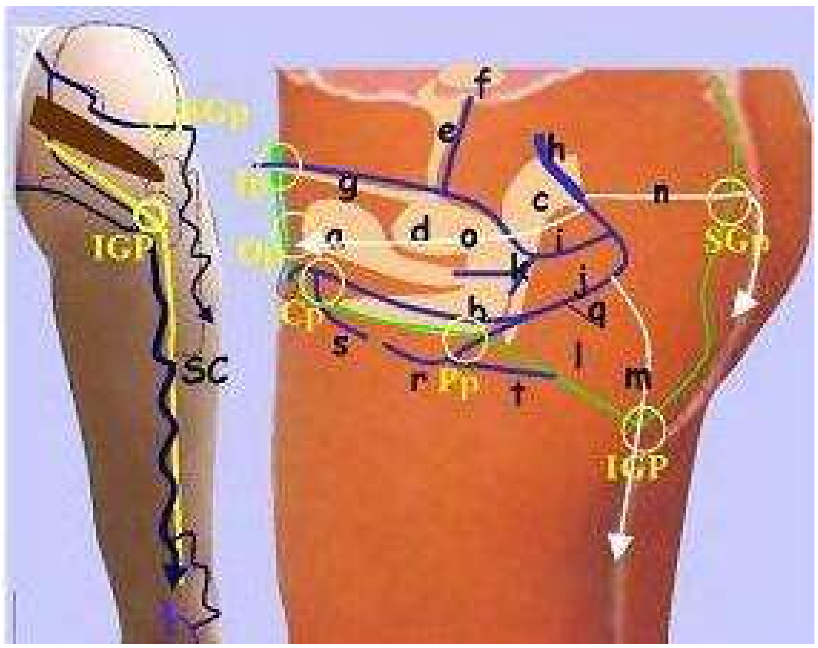

Figure 6.4 B) Three parietal pelvic venous escape points and feeding hypogastric tributaries. a: bladder. b:vagina. c: rectum: uterus. e: fallopian tube + tubo-ovarian vein. f: ovary. g:round ligament of uterus and its vein. h: hypogastric vein. i: uterine vein. j: internal pudendal vein k: vesico-vaginal veins.l: fascia superficialis m: inferior gluteal vein. n:superior gluteal vein. o:obturator vein. P: bulbo-clitoridis vein. q: inferior haemorroidal vein r: labialis posterior vein. s: labialis anterior vein.t: superfificial perineal vein.SC: sciatic nerve and its varicose vein. SGP: Superior gluteal point: reflux location fed by the incompetent superior gluteal vein (n) into a superficial gluteal vein through a fascial perforation. This reflux can feed superficial varices anywhere in the ipsi-lateral limb and connect with any perineal tributary. IGP : Inferior gluteal point: reflux location at the middle of the buttock inferior fold fed by the incompetent inferior gluteal vein (m). This reflux can feed superficial varices anywhere in the ipsi-lateral limb and connect with any perineal tributary but frequently deep varices around the sciatic nerve and then surface in the popliteal region. Op: Obturator point: reflux location at the groin where the obturator vein connects distally with the great saphenous arch or the common femoral vein , fed by the incompetent obturator vein (o) connected proximally whith the hypogastric vein (h) or with the internal pudendal vein. This reflux can feed superficial varices anywhere in the ipsi-lateral limb, but particularly through its coonection with the great saphenous arch. Otherwise, if the distally connection is unique with the common femoral vein, superficial veins cannot be overloaded.

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Chapter 7

THE CHIVA STRATEGY AND OTHER CONSERVATIVE AND HEMODYNAMIC TREATMENTS

Claude Franceschi
Saint Joseph and Pitié-Salpétrière Hospitals, Paris, France.

1. CHIVA DEFINITION

CHIVA is a French acronym for a type of treatment: Conservatrice et Hemodynamique de l'Insuffisance Veineuse en Ambulatoire, which means conservative and hemodynamic treatment of venous insufficiency in ambulatory care [100-106]. The original goal of CHIVA was the preservation of the superficial venous capital for further possible arterial by-pass necessity, thus any aggressive treatment of the superficial venous network was avoided. However, the fact that varices and most other venous disorders are treated by lying down and, above all, by raising the legs suggests that any treatment that can control hydrostatic pressure, including positioning, deserves great medical interest. Most varices and venous insufficiencies are postural because they are triggered by positions in which hydrostatic pressure is too high and not physiologically corrected because of an impairment in dynamic fractionation of hydrostatic pressure (DFHSP). Such impairment is usually caused by superficial and deep closed shunts (CS) or superficial derived open shunts (DOS) due to venous incompetence that disrupts DFHSP control by the valvulo-muscular pump (VMP). Since valve repair or prosthesis is neither easy nor feasible to date, interruption at or immediately under escape points should repair the VMP and suppress pathologic shunt flow. At the same time, these interruptions fractionate permanently and dynamically the column of pressure. Furthermore, and in order to fragment better the hydrostatic column in long incompetent superficial veins, other interruptions immediately under possible competent perforators connected with those veins are performed. This means that CHIVA restores venous physiology in walking patients by maintaining a low distal venous pressure due to renewed VMP efficiency and its effect on DFHSP and blood drainage. Ultimately, venous disorders and varices are supposed to heal. Venous disorders caused by excessive transmural pressure (TMP) should heal due to restoration of DFHSP and venous drainage repair. Varicose calibres should decrease to normal thanks to aggressive shunt flow suppression and efficient VMP blood aspiration. These theoretical hypotheses have been confirmed by numerous publications since their initial presentation at Precy-sous-thil France in October 1988, at which time a text was published entitled Theorie et pratique de la Cure Conservatrice et Hemodynamique de l'Insuffisance Veineuse en Ambulatoire [105].

2. Fundamentals of the CHIVA Strategy

2.1. Strategy, Tactics, and Definitions

Strategy is the intellectual concept of the actions that have to be taken in order to reach a goal according to a theoretical model. Tactics are the material means selected to perform the actions according to the strategy. In this field, successful strategy relies on understanding venous pathophysiology, and the efficiency of the tactics depends on the quality of the material means. For example, an arterial by-pass can fail because of a mistaken strategy (wrong indication according to the arterial pathophysiology knowledge) or because of a bad tactic (faulty prosthetic material or surgical error). Many CHIVA failures are either due to disrespect or misunderstanding of the strategy or inadequate material means. For this reason, both a clear understanding of the theory and adequate training are necessary to perform CHIVA.

2.2. Fundamental Rules of Strategy

CHIVA strategy is conservative not only to preserve the venous capital for further arterial by-pass, but also because vein destruction involves hemodynamic disorders. Actually, venous destruction precludes the drainage that causes tissue suffering and varicose recurrence by vicarious effect. CHIVA strategy is hemodynamic because it restores proper distal venous pressure. As a matter of fact, CHIVA reduces lateral pressure and consequently transmural pressure (TMP) by restoring DFHSP through disconnection of closed shunts, suppressing pathogenic shunt flows, and preserving beneficial vicarious shunts. CHIVA strategy is focused on the re-entry quality, that is, on its ability to drain the venous network properly in order to avoid the effects of drainage preclusion. For this reason, limited venous interruptions are preferable to extensive ones, even if the immediate aesthetic outcomes are less satisfactory. Actually, aesthetic outcomes improve with time, long-term results are much better, and recurrences are much fewer with limited interventions. Clearly, however, any removal or occlusion of nondraining or redundant veins is not contrary to CHIVA strategy, which consists of the following principles:

A Preservation of deep and superficial draining flows even in varicose veins.

B Disconnection of deep and superficial CS and superficial DOS in a way that blocks the shunting flow WITHOUT precluding the draining flow.
C Fractionation of the hydrostatic pressure column WITHOUT precluding the draining flow.
D Adaptation of the re-entry efficiency to the needs of draining flow.
E Minimization of venous interruptions. The less the veins are interrupted, even in extended varicose, the better the long-term results. The more the veins are destroyed, even if varicose, the better the short-term results but the worse the long-term [100-106].

2.2.1. Preservation of Deep and Superficial Draining Flows Even in Varicose Veins

Any venous flow, even in varicose veins, is composed totally or partially of physiologic draining blood from tissues. Any blockage of that draining flow involves upstream tissue suffering including capillaro-venular overload (telangiectasias and micro-varicosities). Collateral veins, overloaded and dilated by the force of draining flow, act as natural venous by-passes that circumvent a blockage and allow for physiologic drainage of the upstream tissues. In this way, a vicarious open shunt (VOS) is formed. VOS can be elicited by a functional block, thrombosis, sclerosis, ligation or removal, the latter two being the reason for recurrent varices after disrespect of draining flows. There is no indication for VOS destruction, even if VOS is an unsightly varice. However, it is obvious that removal or occlusion of non-draining or redundant veins is compatible with CHIVA strategy.

2.2.2. Disconnection of Deep and Superficial Closed Shunts (CS) and Superficial Derived open Shunts (DOS) in a way that Blocks Shunting flow WITHOUT Compromising Draining Flow

2.2.2.1. ***Disconnection of deep closed shunts (CS).*** The principle of CHIVA in deep veins is disconnection of the shunting deep incompetent vein at its junction with the deep competent but shunted vein. Thus, during VMP systole, the distal flow is totally injected into the competent vein and does not flow back again during the diastole. Also, correct dynamic fractionation of hydrostatic pressure (DFHSP) is once again possible. The competent shunted vein can be close to or distant from the incompetent shunting. It is sufficient that the blood drained by the incompetent vein can be transported by the competent one. If it cannot be, disconnection of the incompetent deep vein is obviously contraindicated because it would be hazardous to physiologic drainage. For example, the incompetent collateral of a double superficial femoral vein can be successfully blocked if the other one is competent. An incompetent superficial femoral vein can be blocked only if the deep femoral vein is competent and participates to the drainage of the calf. The same reasoning applies to deep shunts of the calf. An incompetent posterior tibial vein, for example, can be interrupted if the competent fibular vein can drain the tibial territory and vice versa.

2.2.2.2. ***Disconnection of superficial closed shunts (CS).*** Superficial veins can shunt other superficial or deep veins in closed circuit either open circuits.

2.2.2.2.1. Superficial veins shunting other superficial veins in closed circuit. This type of shunt is usually not much overloaded because it is not connected directly with the deep network, so it is not much activated by the VMP. However, its disconnection at the escape junction can reduce its calibre. Furthermore, this shunting vein can be destroyed when redundant. It corresponds to shunt type 2.

2.2.2.2.2. Superficial veins shunting deep veins in closed circuit. These represent a major cause of chronic venous insufficiency and varicose veins. They are classified in shunts type 1, 3, 4, 5, and 6 according to the patterns of the superficial veins that shunt the deep veins. CHIVA strategy attempts to disconnect the closed shunts at the escape point without affecting the physiological blood drained by the shunting vein. Therefore, particularities of disconnections depend on the hemodynamic configuration of each CS. Disconnection at the escape point (i.e., at the N1-N2 junction) has to take into account the shunt's ability to provide for efficient re-entry for the proximal tributaries. If it can, N2 can be disconnected immediately beneath the junction. If not, N2 will be disconnected at the junction.

2.2.2.2.2.1. Shunt 1 CHIVA strategy: Shunt 1 is N1-N2-N1. It can involve the great or small saphenous veins. One has to make sure that these shunts are not just the diastolic phase of a mixed shunt because the strategy would be different.

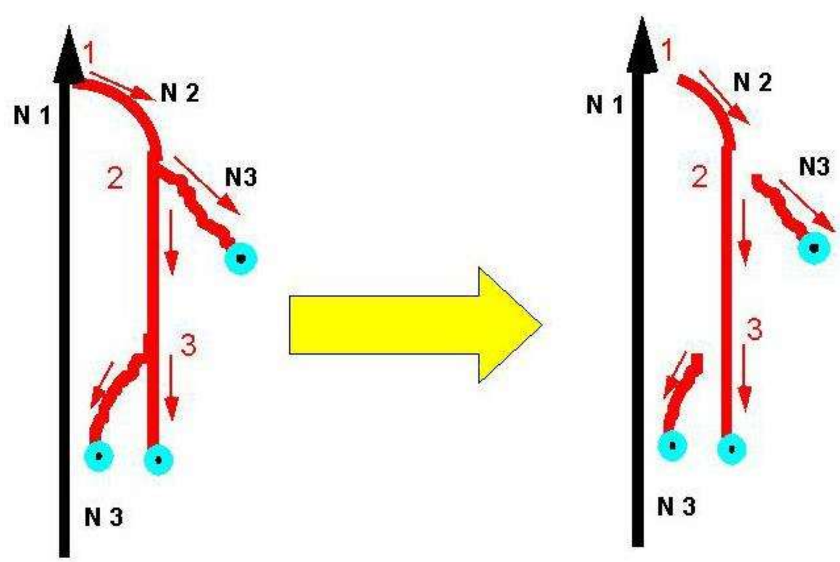

Figure 7.1. Shunt type 1+N3 and associated CHIVA 1 procedure.

2.2.2.2.2.1.a. Shunt 1 in great saphenous vein (GSV): When the sapheno-femoral junction (SFJ) is the escape point (EP), GSV is precisely disconnected at the GSV. This disconnection could be made immediately beneath the arch in order to drain the arch and its tributaries through the SFJ. This is not because of the risk of recurrence by the arch tributaries. As a matter of fact, at this point aspiration is weak (no VMP), and terminal valve incompetence does not prevent excessive pressure in the deep network (from coughing, defecating, lifting heavy loads, etc.). For these reasons, the arch and its tributaries are better drained by distal re-entry.

When the escape point is an incompetent perforator connected to the saphenous trunk, it remains preferable to disconnect the perforator itself when the corresponding VMP is not efficient. This is the case in the thigh. On the contrary, in the lower region of the leg, disconnection can be performed at the saphenous trunk immediately beneath the incompetent perforator thanks to the efficiency of calf VMP. This efficiency can be assessed by Doppler, which shows whether or not there is an inflow during VMP diastole while N2 is compressed by a tourniquet or a finger.

2.2.2.2.2.1.b. Shunt 1 in small saphenous vein (SSV): Usually the RP is at the saphena-popliteal junction (SPJ).Theoretically, the shunting SSV has to be disconnected at SPJ. In fact, Giacomini vein (GV) hemodynamics allow disconnecting of the SSV immediately beneath its junction with GV because GV can function as a vicarious pathway in case of flow-pressure excess in the popliteal vein, thereby preventing recurrences through forced perforators of popliteal fossa.

2.2.2.2.2.2. Shunt 3 CHIVA strategy: Shunt 3 is N1-N2-N3-N1. It can involve the great or small saphenous veins. One has to make sure that these shunts are not just the diastolic phase of a mixed shunt because the strategy would be different. Shunt 3 differs from shunt 1 by N3 segment interposition between N2 and the re-entry point (REP). The strategies are also different. Actually, disconnection at the junction as in shunt 1 is problematic because it would disconnect a closed shunt at N1-N2 but leave a derived open shunt (DOS) N2-N3 called shunt 2. N3 would be less overloaded because CS shunting flow would have been interrupted and the pressure column would have been fractionated at the arch, but overload would still be excessive because of the remaining shunt 2. Disconnection at both N1-N2 and N2-N3 junctions would disconnect both shunts 1 and 2, but N2 draining flow would be precluded because there would be no available re-entry points (REP) on its track. Even if this approach is conservative, disconnects the shunts, and fractionates the hydrostatic pressure, double disconnection in shunt 3 is not CHIVA because it does not allow draining flow. In order to carry out a correct CHIVA in shunt 3, two different procedures are possible. The first is limited to one disconnection at N2-N3 that stops the shunt 1 and 2 shunt flows. In that case, N2 and N3 drain no more shunt flows. N3 is retrograde but drains only its territory, and hydrostatic pressure is fractionated at the N2-N3 junction. N2 is antegrade due to no efficient *re-entry* on its track, but it is still subjected to the previous nonfractioned column of hydrostatic pressure because of no N1-N2 fractioning. N2 segment submitted to excessive hydrostatic pressure explains two phenomena: 1) calibre is partially reduced due to shunt flow suppression but does not return completely to normal, and 2) in some cases a previously small and inefficient perforator located on this N2 segment can enlarge, so that N2 flow is reversed and the previous shunt 3 is converted to shunt 1. In the latter case, the induced shunt has to be corrected, like any shunt 1, by a disconnection at the N1-N2 junction. This procedure is called CHIVA 2 because it may need of two different procedures planned when the duplex investigation confirms the transformation of type 3 shunt into type 1. The latter consists of disconnections at both the N1-N2 and N2-N3 junctions associated with distal N2 devalvulation until an efficient re-entry is represented by one or more perforators.

2.2.2.2.2.3. Shunt 4 CHIVA strategy: Shunt 4 is a CS N1-N3-N2-N1. EP at the N1-N3 junction is usually a pelvic escape point, gluteal, obturatory, inguinal (I point) or perineal (P point). Disconnection must be performed only at the EP.

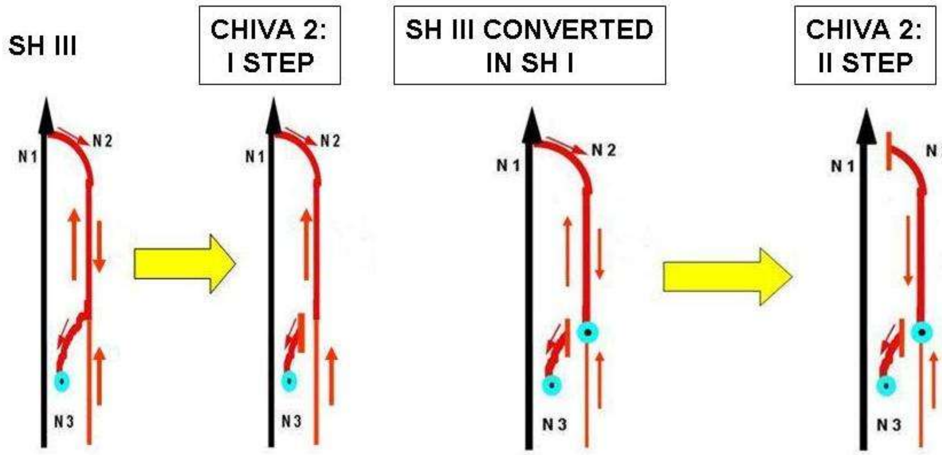

Figure 7.2. CHIVA II first and second step for Shunt type III.

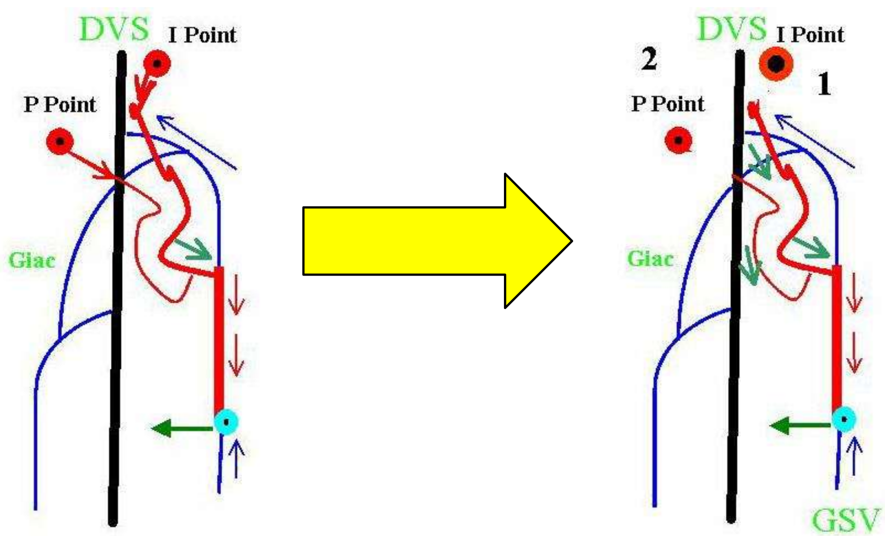

Figure 7.3. Shunt type IV and associated CHIVA procedure.

2.2.2.2.2.4. Shunt 5 CHIVA strategy: Shunt 5 is a CS N1-N3-N2-N3-N1. As in shunt 4, EP at the N1-N3 junction is usually a pelvic escape point, gluteal, obturatory, inguinal (I point) or perineal (P point). Disconnection must be performed not only at the N1-N3 EP, but also at the N2-N3 in order not to leave a shunt 2.

2.2.2.2.2.5. Shunt 6 CHIVA strategy: Shunt 6 is a CS N1-N3-N1. EP can be a pelvic escape point or any other EP. This shunt can appear in congenital malformations but is more usually seen in recurrences following removal of varices and saphenas.

2.2.2.3. ***Disconnection of a mixed shunt (MS):*** A mixed shunt is composed of two shunts: a vicarious open shunt (VOS) and a closed shunt (CS) that share the same EP and distal segments of the shunting veins. Their REP and proximal segments are different. REP of VOS is proximal to LP, and REP of CS is distal to LP. As VOS has to be preserved and CS disconnected, the disconnection cannot be at LP but only at the distal venous segment of CS where it diverges from the distal venous segment of VOS.

2.2.2.4. ***Disconnection of superficial derived open shunts (DOS):*** DOS are shunts 2, N2-N3-N1. Disconnection of DOS is done simply on N3 at the N2-N3 junction.

2.2.2.5. ***Disconnection of composite shunts:*** Practically speaking, most shunts are composite in that a shunt 1 can be connected to a shunt 2 and so on. Sometimes the REP of one shunt is not efficient enough, and drainage is also provided by the REP of an associated shunt. Therefore, any shunt can be disconnected as long as its disconnection does not preclude any draining flow in the associated shunt. For this reason, the proper REP of every associated shunt has to be assessed by dynamic tests. For example, when a composite shunt is shunt 1+2 (N1-N2-N1-N2-N3-N1), disconnection can be carried out at N1-N2 and N3-N1 as long as REP in N2 is efficient. If REP in shunt 1 is not obviously efficient, this composite shunt can be assimilated to a shunt 3 and treated by CHIVA 2 or CHIVA devalvulation. In another case, two EPs can be superimposed and be drained by the same distal REP (N1-N2-N1-N2-N1). Fractionation can be done immediately at the upper EP. It may be also done beyond the lower EP even if refluxing during diastole, but only if testing shows it to be efficient enough as REP. Thus, under appropriate conditions, a previous EP can be transformed into REP.

2.2.3. Static Fractionation of Hydrostatic Pressure Column WITHOUT Compromising Drainage

Disconnection of CS and DOS shunts also allows for static fragmentation of the column of hydrostatic pressure. Sometimes the incompetent segment disconnected from the shunt is too long and high, so it needs a complementary fractionation. As this fractionation must not preclude drainage, every remaining subsegment of vein must have a REP.

2.2.4. Adaptation of the Re-Entry Efficiency to the needs of Draining Flow

Re-entry is the place where shunt flow rejoins its physiologic pathway. After CS or DOS disconnection, flow is limited to the draining flow but can be too great for the capacity of the REP, either because its calibre is too small or because downstream aspiration is too weak. REP is efficient when it allows a correct drainage of the network that converges on it, that is, when its calibre is sufficient and downstream aspiration is correct.

When more than one REP drain a CS or DOS, hemodynamic assessment of each perforator is performed by diastolic flow velocity registration in the shunting vein while other veins that carry the same flow to other re-entries are blocked. An inefficient re-entry leads to tissue suffering, telangiecstases, and vicarious varices.

2.2.5. Superficial CS and Deep Venous Incompetence

In cases of superficial CS or DOS, the greater the efficiency of the VMP, the more the diastolic flow. On the contrary, the lower the VMP efficiency, the less the diastolic flow. That is why, in cases of totally inefficient VMP, as when the deep veins are totally incompetent, dynamic tests of VMP will not show any diastolic flow in the superficial veins, even if they are very large and varicose. Actually, during the diastole, total incompetence does not allow the VMP to reduce venous pressure, so the gradient of the re-entry perforating vein is not favorable to inflow. This is illustrated by the Perth test, which shows no reduction of varices when walking despite a tourniquet at the hip. It is also demonstrated by an unexpected lack of reflux in superficial varicose veins during dynamic tests, such as Paranà maneuver. The CHIVA strategy is to correct before deep incompetence occurs, through disconnection in case of deep closed shunt or valve repair if there is total incompetence without any CS, followed by disconnection of superficial shunts.

Superficial disconnection without deep incompetence correction makes no haemodynamic sense because no HDSFP cannot be achieved

2.3. Post Chiva Controls

After CHIVA, clinical and instrumental tests have to be interpreted according to CHIVA patterns.

2.3.1. Regression of Clinical Symptoms: is favored by walking. All functional symptoms and oedemas related to venous insufficiency, such as postural pains and heaviness, should disappear or diminish during the first postoperative days. Ulcers should close within few weeks. Hypodermitis should disappear in the course of several weeks. Pigmentation should regress after several months, up to two years. The caliber of varicose veins should return to normal within a few days or weeks, depending upon their pre-operative size.Thus, clinical outcome cannot be definitely assessed until some weeks or months after the procedure.

2.3.2. Post-CHIVA Echo-Doppler Assessments: Valsalva maneuver has to be negative at the operated shunts. Retrograde flow during dynamic tests does not necessarily signify a failure. Poor systolic but significant diastolic retrograde flow attests to shunt 0, thus a correct drainage [6,45,47,56,255].

2.3.3. Air PLethysmography Assessments: The pressure-volume relationship in the leg has to normalize according to the efficiency of the procedure [251].

2.4. Post-Chiva Iterative Procedures

Except for CHIVA 2, no further procedures usually have to be performed. Actually, in complex cases such as multi-shunts in venous congenital malformations, it is better to spread the cure over two or more operations every six months in order to avoid excessive interruptions and to clarify the situation progressively.

2.5. Short and Long Term Results

The less the veins are interrupted, even in extended varicose, the better the long-term results [56]. The more the veins are destroyed, even if they are varicose, the better the short-term results but the worse the long-term. It is preferable to spread CHIVA over two distant procedures, thereby delaying to the second operation those complementary interruptions that are not obviously safe for drainage. This is the case, for example, when in a shunt 1+2, the REP in shunt 1 is not obviously efficient. This composite shunt can be assimilated to a shunt 3 and treated by CHIVA 2. These iterative operations, when necessary, are not difficult because they are ambulatory, brief, and performed using local anaesthesia.

Obviously, complete removal or closure of all varicose veins results in immediate aesthetic success. Telangiectases, micro-varicosities and vicarious varices that appear over time are not due to a "natural" evolution of the varicose disease, but rather to long-term complications of nonhemodynamic procedures. On the contrary, hemodynamic procedures like CHIVA preserve the veins, even if varicose, in order not only to spare the venous capital for further arterial by-pass, but also to restore correct venous function. The so-obtained correct venous function progressively cures varicose veins and trophic disorders and prevents further recurrence. The time necessary for healing ranges from several days to several weeks, and the varicose veins remain visible during that period. Patients accept this delay for a better final outcome.

2.5.1. Chiva and Recurrence

Varicose recurrence is usually imputed either to an incomplete destruction of superficial veins (especially incomplete resection of the arch of great saphena or small saphena and its tributaries) or to the natural evolution of varicose disease. Actually, thanks to pre- and post-operative echo-Doppler findings, the problem can be identified as one of four conditions.

Same varices reappear: Varicose reappearance over time (of the same varices as before the therapy) constitutes true recurrence. An example is when an occluded or caliber- reduced varicose vein becomes permeable and dilated again. In this case, recurrence could be due to the forcing of veins by draining flow because of inefficient re-entry or an inefficient interruption. If the recurrence is due to an inefficient REP, it is better not to repeat any interruption at the same point, but rather check another one beyond a better REP. If the recurrence is due to a tactical (material) interruption failure, a more reliable technique is necessary.

Remaining varices: Some or all of the varices may be left untreated by the therapist e.g., when some or all of varices remain immediately after therapy. In this case, a complementary CHIVA can be performed.

Varices in different territories: Varices different from those previous to therapy can be due to an independent varicose evolution, e.g., when new varices of previously normal small saphena occur after therapy of great saphena. Here, too, complementary CHIVA can be carried out.

Iatrogenic varicose veins: Varices may be induced by previous therapy, so that one can call them long-term iatrogenic complications. For example, it is not rare to notice new varices without any leak point but made only of vicarious veins. This occurs because previous draining normal or varicose veins were destroyed. In the case of superficial VOS due to drainage preclusion, static fractionation is possible only if intermediate perforators (REP) are found on their track. In the case of CS created by anarchical VOS that open incompetent perforators, CHIVA has to disconnect them. Unfortunately, most cases of recurrence after venous removal are VOS without much intermediate REP, which reduces the therapeutic possibilities.

3. Particular CHIVA Strategies

When venous pressure increases, transmural pressure (TMP) increases so that liquids and metabolic wastes from the tissues cannot pass into circulation. An obstacle to the passage of liquids involves the appearance of oedemas. Intra-tissue accumulation of toxic metabolites associated with capillary flow slowdown is responsible for trophic disorders. The reactive vasodilation of arterioles and the opening of micro-shunts worsen tissue ischemia in two ways: 1) TMP increases because of residual pressure (RP) enhancement; and 2) the opening of micro-shunts steals capillary flow, thereby worsening cellular necrosis. This explains the coexistence of oxygenated (red) blood with necroses in venous ulcers. Infection occurs because of the ideal culture medium that this type of ulcer represents. CHIVA has to reduce TMP through lateral pressure reduction. Disconnection of shunts and fractionation of pressure have to be done with the utmost attention to drainage of the damaged tissue. However, those sites that theoretically have to be interrupted but that involve inflamed tissues must not be touched at the first operation. The first procedure is performed only at sites having healthy tissues. A second procedure is performed at the remaining sites when their trophicity is improved by the first procedure. Actually, correct and long-term durable interruptions cannot be obtained in inflamed tissues.

3.1. Congenital Venous Malformations

Examination is necessary to identify which varicose and/or incompetent deep and/or superficial veins are CS, DOS, VOS, mixed shunts, or NON draining malformative vein. In these multiple and complex situations where, in addition to shunting flows, superficial "physiologic" flows are increased by superficial angiomas, interruptions must divided into multiple operations in order to correct hemodynamic disorders in the most efficient, accurate, and safe manner.

4. CHIVA MAPPING

Mapping is a schematic design that depicts the venous anatomo-functional configuration of each patient. Flow direction according to the *systole-diastolic* phase of dynamic tests and Valsalva maneuver is indicated by an arrow for each depicted vein. Data are obtained through echo-Doppler. In this way EP, shunting veins, REP, and occluded or removed veins can be identified and analyzed in order to plan an appropriate CHIVA strategy. Only echo-Doppler allows such mapping and thus a correct CHIVA strategy.

Clinical examination is not sufficient to assess accurately different types of shunts. Use of an ultrasound duplex scanner is mandatory because it is the only noninvasive instrument able to show in real time the anatomy of the venous network and, contemporaneously, venous flow direction and velocity in any position and during any dynamic maneuver.

4.1. Valsalva Manoeuver (VM)

Positive VM, that is an antegrade or retrograde flow elicited by thoraco-abdominal and diaphragm contraction, evidences a deep incompetence when in deep veins and a closed shunt CS when in superficial veins. Escape point is checked upstream. Valsalva maneuver is negative in VOS and DOS [46,48,241].

4.2. Manual or Pneumatic Squeezing of the Calf

It can evidence reflux upon relaxation but can be false negative when squeezing is performed incorrectly or when the aspiring VMP sector is not involved by compression. Moreover, squeezing compresses superficial veins, which does not occur during physiologic VMP motion [102,148,150].

4.3. Dynamic VMP Activation

Paranà maneuver allows a “physiologic contraction” and, at the same time, facilitates ultrasound scanning in the standing immobile position due to an isometric reflex contraction of the muscles of the lower limbs to a slight push-pull at the patient’s waist. Flow is activated by VMP systole and /or diastole [102].

4.3.1. In the superficial network, diastolic flows are elicited by the VMP in CS and DOS. Systolic flows are elicited in CS, VOS and DOS. Diastolic velocity is proportional to the shunting flow. Diastolic and systolic flow directions are usually opposite in CS and

DOS but can be the same *as* CS that involve Giacomini vein. In the case of multiple REPs in CS or DOS, the efficiency of the REPs proximal to the terminal can be assessed by diastolic flow velocity while compressing the terminal re-entry. This assessment is useful to the hemodynamic therapeutic strategy. Reflux time, Psatakis index, or dynamic reflux index (DRI) measurement can evidence reflux only when systolic and diastolic flow directions are opposite. Flow directions remain pathologic even when pressure and velocity return to normal but remain inverted after shunt disconnection. In this case, flow direction is abnormal but no longer pathogenic. In some cases, the flow in great incompetent varicose saphena cannot be activated by diastole when the VMP is disabled because of total deep venous valve incompetence. That is why valve incompetence allows retrograde flow only if the pressure gradient is inverted, that is, when the VMP is efficient. In other cases, diastolic reflux occurs while deep venous veins are refluxing, too. This means that there remains an efficient valvulo-muscular portion connected to the shunt. In other words, the greater the superficial diastolic reflux, the better the VMP efficiency.

4.3.2. In the deep network, systolic and diastolic flows are elicited by the VMP. Diastolic flow evidences a proximal valve incompetence without deep CS when diastolic flow volume is less than or equal to systolic, or when DRI ≥ 1. Deep CS is demonstrated when diastolic flow volume is greater than systolic, or DRI >1, because, as explained previously, CS carries deviated flow in addition to normal during diastole. Detection of deep CS is the central goal of deep venous incompetence investigation because it is determinant for hemodynanamic therapeutic strategy.

5. Other Conservative and Hemodynamic Treatments

CHIVA can be associated with other hemodynamic and conservative techniques but is incompatible with nonhemodynamic techniques [43,98,99,126-128,153,154]. In order to reduce transmural pressure (TMP), those methods affect lateral or external pressure. Surgical and medical treatments can be classified into two categories, according to whether they are in accordance with hemodynamic principles or not. TMP is the central parameter of the regulation of venous function. Consequently, treatments must restore a TMP that is favorable to drainage and an optimal venous capacity. The therapeutic action must consist of either increasing extra-venous pressure or decreasing intra-venous pressure.

5.1. Increase in Extra Venous Pressure

An increase in external pressure reduces TMP and thus supports drainage and the reduction of venous caliber. This can be obtained by an increase in atmospheric pressure, which is maximal at sea level. Generally, however, increases in EP are obtained through passive external compression with rigid or elastic bandages, but also by immersion in a water bath or even in a mercury bath. Sometimes active pneumatic compression is used. In all cases, the goal is reached when the TMP is corrected. Such applied external pressure cannot exceed the arterial blood pressure because of the risk of ischemia. Also, when intravenous pressure is very high, strong external compression can be painful due to the crushing of tissues between excessive intravenous and external pressures. Therefore, external compression is appropriate only when the intravenous pressure is not too high. The hemodynamic effects of rigid and elastic bandages differ in two ways. Rigid application increases VMP efficiency during systole because it confines the content of the pump in a rigid container, but it is less efficient at rest. Elastic application cushions the hemodynamic effects of the systole and is more efficient at rest. For these reasons, rigid application is preferred for subjects who walk. In any case, compression represents a logical and effective hemodynamic treatment for venous insufficiency. It is free of side effects except when overdone, and it can be applied alone or associated with other treatments. Because of its direct effect on TMP, compression also constitutes also a major preventive treatment under temporary physiological conditions, such as presence at high altitude, long-term standing or sitting, and pregnancy.

5.2. Reduction of Venous Pressure

Reduction of venous lateral pressure (LP) decreases TMP, thereby improving drainage, venous size, and venous caliber. LP depends on HSP and the lateral component of residual pressure (RP).

5.2.1. Postural Therapy:

At the ankle, HSP is maximal in the motionless standing position, low in the lying position, and negative (lower than the atmospheric pressure) when the foot is raised above head level. Therefore, posture can control TMP by action on the HSP. So, lying and, moreover, raising lower limbs support drainage and decrease venous caliber.

5.2.2. Thermo-Therapy:

Heat increases RP by reducing microcirculatory resistance and increases LP as well. Thus, cooling the legs reduces LP.

5.2.3. Valve Repair or Valve Prosthesis:

In cases of deep venous valvular incompetence without any shunt, the theoretical treatment consists of valvular repair or valvular prosthesis [142,168,265-267]. Unfortunately, these techniques have need of improvement. In most cases, only postural and compressive therapy can be applied.

5.3. External and Lateral Pressure-Associated Correction

Treatments for venous insufficiency by TMP correction often combine external pressure increase with LP reduction. This occurs when one walks into the sea because the external effects of compression by water are combined with maximal atmospheric pressure, with LP reduction caused by reduced superficial flow from thermoregulation, and with DFHSP induced by walking. It is also the case when an imperfect correction of closed shunts is supplemented by compression.

6. Tactic Material Means in CHIVA

Which are the best procedures to achieve the strategic decisions? In other words, which are the most accurate, efficient and lasting material means to perform flow interruptions at the right places according to strategic needs? How not to confuse tactic failure with strategic error? Is a forced or by-passed interruption due to an inappropriate tactic or a mistaken strategy?

6.1. Surgery

First of all, a previous precise marking under duplex ultrasound by an operator aware of the surgical necessities is indispensable. Venous short resection (1 to 4 cm) associated with nonresorbable ligation and nonresorbable closure of the perforated fascia seems to be the most precise, efficient, and long-lasting material means to date. Simple nonresorbable ligations are seldom forced and reopened. Multiple ligations with nonresorbable thick thread seem to resist better. Resorbable venous ligation after section could favor a recanalisation due to inflammatory angiogenetic effects.

6.2. Endovenous Procedures:

Endovenous procedures, such as sclerotherapy, laser, and radiofrequency, are not yet appropriated for precise, limited, and lasting interruptions. Nevertheless, they are useful when surgery is not feasible, as for example in sciatic nerve varices where surgery is not suitable for the great risk of iatrogenic nerve damage [199].

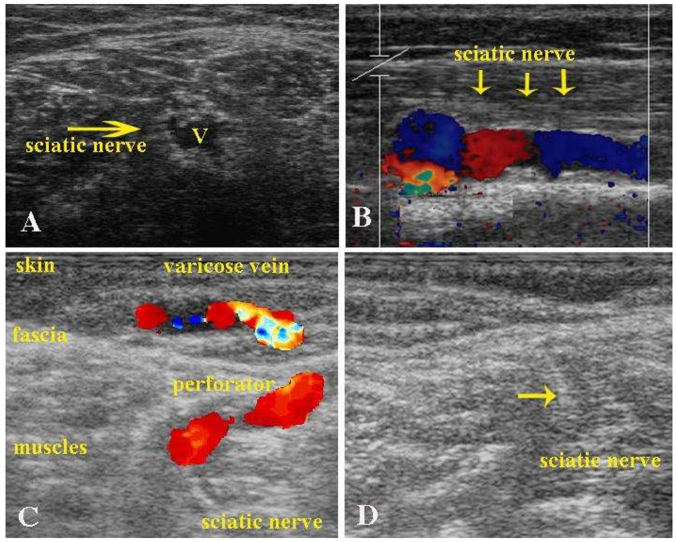

Figure 7.4. Sciatic nerve varices and its therapy by echo-guided sclerotherapy. The sciatic veins are deep veins and lay within the sciatic nerve The nerve was easily visible on ultrasound imaging because of thickening of the neural sheath and dissection of the nerve fibres by the dilated vessels (Fig. A transversal access, B longitudinal access. C varicose veins ot the lateral aspect of the leg fed by reflux coming from sciatic veins through a perforator Please note in D) the echo coming from the foam reaching its target in an area not so easily reachable by surgery.